Interpolation and the Lagrange polynomial

In this chapter, we discuss the problem of approximating a given function by polynomials. There are two main uses of these approximating polynomials. The first use is to reconstruct the function $f(x)$ when it is not given explicitly and only values of $f(x)$ and/ or its certain order derivatives are given at a set of distinct points called *nodes* or *tabular points*. The second use is to perform the required operations which were intended for $f(x)$, like determination of roots, differentiation and integration etc. can be carried out using the approximating polynomial $P(x)$. The approximating polynomial $P(x)$ can be used to predict the value of $f(x)$ at a non-tabular point. The deviation of $P(x)$ from $f(x)$, that is $f(x) - P(x)$, is called the *error of approximation*.

Let $f(x)$ be a continuous function defined on some interval $[a, b]$, and be prescribed at $n + 1$ distinct tabular points $x_0, x_1,..., x_n$ such that $a = x_0 < x_1 < x_2 < ... < x_n = b$. The distinct tabular points $x_0, x_1,..., x_n$ may be non-equispaced or equispaced, that is $x_{k+1} - x_k = h$, $k = 0, 1, 2,..., n - 1$. The problem of polynomial approximation is to find a polynomial $P_n(x)$, of degree $\leq n$, which fits the given data exactly, that is,

$$P_n(x_i) = f(x_i), \quad i = 0, 1, 2,..., n. \tag{1}$$

The polynomial $P_n(x)$ is called the *interpolating polynomial*. The conditions given in (1) are called the *interpolating conditions*.

Lagrange Interpolation

Let the data

x	x_0	x_1	x_2	...	x_n
$f(x)$	$f(x_0)$	$f(x_1)$	$f(x_2)$	...	$f(x_n)$

be given at distinct unevenly spaced points or non-uniform points $x_0, x_1,..., x_n$. This data may also be given at evenly spaced points.

For this data, we can fit a unique polynomial of degree $\leq n$. Since the interpolating polynomial must use all the ordinates $f(x_0), f(x_1),... f(x_n)$, it can be written as a linear combination of these ordinates. That is, we can write the polynomial as

$$\begin{aligned} P_n(x) &= l_0(x) f(x_0) + l_1(x) f(x_1) + ... + l_n(x) f(x_n) \\ &= l_0(x) f_0 + l_1(x) f_1 + ... + l_n(x) f(x_n) \end{aligned} \tag{2}$$

where $f(x_i) = f_i$ and $l_i(x)$, $i = 0, 1, 2, ...,n$ are polynomials of degree n. This polynomial fits the data given in (1) exactly.

At $x = x_0$, we get

$$f(x_0) \equiv P_n(x_0) = l_0(x_0) f(x_0) + l_1(x_0) f(x_1) + ... + l_n(x_0) f(x_n).$$

This equation is satisfied only when $l_0(x_0) = 1$ and $l_i(x_0) = 0$, $i \neq 0$.

At a general point $x = x_i$, we get

$$f(x_i) \equiv P_n(x_i) = l_0(x_i)\, f(x_0) + \ldots + l_i(x_i)\, f(x_i) + \ldots + l_n(x_i)\, f(x_n).$$

This equation is satisfied only when $l_i(x_i) = 1$ and $l_j(x_i) = 0$, $i \neq j$.

Therefore, $l_i(x)$, which are polynomials of degree n, satisfy the conditions

$$l_i(x_j) = \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases}. \qquad (3)$$

We seek a polynomial of degree $\leq n$ which at all the x_i with exception $x_k\,(x_i \neq x_n)$

Let $g_k(x) = \prod_{i=0, i\neq k}^{n} (x - x_i) = (x-x_0)(x-x_1)(x-x_2)\cdots(x-x_{k-1})(x-x_{k+1})\cdots(x-x_n)$

- $g_k(x_k) = \prod_{i=0, i\neq k}^{n} (x_k - x_i) = (x_k-x_0)(x_k-x_1)\cdots(x_k-x_{k-1})(x_k-x_{k+1})\cdots(x_k-x_n) \neq 0$

$L_k(x) = \dfrac{g_k(x)}{g_k(x_k)}, x_k \neq 0$ (Lagrange function)

$$= \prod_{i=0, i\neq k}^{n} (x - x_i) \Big/ \prod_{i=0, i\neq k}^{n} (x_k - x_i) = \prod_{i=0, i\neq k}^{n} \frac{(x - x_i)}{(x_k - x_i)}, \qquad L_k(x_i) = \begin{cases} 1, i = k \\ 0, i \neq k \end{cases}$$

For any arbitrary function f(x),

Define $p(x) = \sum_{k=0}^{n} f(x_k) l_k(x)$ ----------(1) (Lagrange form)

$L_k = \prod_{i=0, i\neq k}^{n} \dfrac{(x - x_i)}{(x_k - x_i)} \Rightarrow p(x) = \sum_{k=0}^{n} \prod_{i=0, i\neq k}^{n} f(x_k) \dfrac{(x - x_i)}{(x_k - x_i)}$ is a polynomial of degree $\leq$ n and interpolate f(x) at $x_0, x_1, \ldots, x_n$.

Let $n = 2 \Rightarrow x_0, x_1, x_2 \Rightarrow 3$

$$L_0(x) = \prod_{i=0, i\neq 0}^{2} \frac{(x - x_i)}{(x_0 - x_i)} = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}, L_1(x) = \prod_{i=0, i\neq 1}^{2} \frac{(x - x_i)}{(x_1 - x_i)} = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)},$$

$$L_2(x) = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}, p(x) \cong f(x_0) l_0(x) + f(x_1) l_1(x) + f(x_2) l_2(x)$$

Ex (1): (x, f(x)) = {(0,0), (1,1), (2,4)}, find f(1.5) using Lagrange function

$$n=2,(x_0,f(x_0)),(x_1,f(x_1)),(x_2,f(x_2))\equiv(0,0),(1,1),(2,4)$$

$$p(x)=\sum_{k=0}^{2}\prod_{i=0,i\neq k}^{2}f(x_k)\frac{(x-x_i)}{(x_k-x_i)}=f(x_0)\frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}+f(x_1)\frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)}$$

$$+f(x_2)\frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}$$

$$p(x)\cong 0+1\frac{(x-0)(x-2)}{(1)(-1)}+4\frac{(x-0)(x-1)}{(2)(1)}=\frac{x^2-2x}{-1}+2(x^2-x)=-x^2+2x+2x^2$$

$$-2x=x^2$$

$$\therefore p(x)=f(x)=x^2\Rightarrow f(1.5)=(1.5)^2=2.25$$

Ex (2): By Lagrange formula find f(3), f(5).

x	0	1	2	4
f(x)	1	1	2	5

$$p_3(x)=f(x)=\sum_{i=0}^{3}l_i(x)f(x_i)=l_0(x)f(x_0)+l_1(x)f(x_1)+l_2(x)f(x_2)+l_3(x)f(x_3)$$

$$f(3)=\frac{(3-1)(3-2)(3-4)}{(0-1)(0-2)(0-4)}(1)+\frac{(3-0)(3-2)(3-4)}{(1-0)(1-2)(1-4)}(1)+\frac{(3-0)(3-1)(3-4)}{(2-0)(2-1)(2-4)}(2)$$

$$+\frac{(3-0)(3-1)(3-2)}{(4-0)(4-1)(4-2)}(5)=\frac{-2}{-8}+\frac{-3}{3}+\frac{-6}{-4}\cdot 2+\frac{6}{24}\cdot 5=0.25-1+3+1.25=3.5$$

$$f(5)=\frac{(5-1)(5-2)(5-4)}{(0-1)(0-2)(0-4)}(1)+\frac{(5-0)(5-2)(5-4)}{(1-0)(1-2)(1-4)}(1)+\frac{(5-0)(5-1)(5-4)}{(2-0)(2-1)(2-4)}(2)$$

$$+\frac{(5-0)(5-1)(5-2)}{(4-0)(4-1)(4-2)}(5)=\frac{12}{-8}+\frac{15}{3}+\frac{20}{-4}\cdot 2+\frac{60}{24}\cdot 5=-1.5+5-10+12.5=6$$

Ex (3): Using the numbers $x_0 = 2, x_1 = 2.5, x_2 = 4$ to find the second-degree interpolating polynomial for $f(x) = \frac{1}{x}$.

x	2	2.5	4
f(x)	0.5	0.4	0.25

$$p_2(x) = l_0(x)f(x_0) + l_1(x)f(x_1) + l_2(x)f(x_2)$$

$$l_0(x) = \frac{(x-2.5)(x-4)}{(2-2.5)(2-4)} = \frac{x^2 - 6.5x + 10}{(-0.5)(-2)} = x^2 - 6.5x + 10$$

$$l_1(x) = \frac{(x-2)(x-4)}{(2.5-2)(2.5-4)} = \frac{x^2 - 6x + 8}{(0.5)(-1.5)} = \frac{-1}{3}(4x^2 - 24x + 32)$$

$$l_2(x) = \frac{(x-2)(x-2.5)}{(4-2)(4-2.5)} = \frac{x^2 - 4.5x + 5}{(2)(1.5)} = \frac{1}{3}(x^2 - 4.5x + 5)$$

$$\because f(x) = l_0(x)f(x_0) + l_1(x)f(x_1) + l_2(x)f(x_2)$$

$$= 0.5(x^2 - 6.5x + 10) + \frac{0.4}{3}(-4x^2 + 24x - 32) + \frac{0.25}{3}(x^2 - 4.5x + 5)$$

$$= 0.05x^2 - 0.425x + 1.15$$